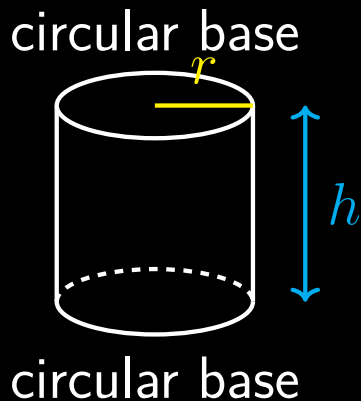


Derivation of the Volume Formula for Cylinders

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A **cylinder** is a three-dimensional figure with two congruent and parallel **circular bases**.

Parts of a Cylinder



$r = \text{radius}$ $h = \text{height}$

Base Area

The **base area** of a cylinder is the area of a circle, given by

$$A = \pi r^2$$

For example, if $r = 3$ cm:

$$A = \pi(3)^2 = 9\pi \text{ cm}^2$$

Height

The **height** is the perpendicular distance between the two circular bases.

Measure it straight from one base to the other, at a right angle (90°).

Stacking Circles



$$A = \pi r^2$$

Imagine stacking many congruent circles. Each circle has an area of πr^2 .

Base Area $\times$ Height

Stacking the circles up to the height h fills the cylinder.

So the **volume** equals the **area of the base** multiplied by the **height**.

$$V = \pi r^2 \times h$$

Volume Formula

The formula for the volume of a cylinder is

$$V = \pi r^2 h$$

where r is the radius and h is the height.

Memory Tip

Think of a stack of coins or pancakes.

$\text{Volume} = \text{Area of one circle} \times \text{Height of the stack}$

Units of Volume

The units of volume are expressed in **cubic units**.

Length	Volume
cm	cm^3
m	m^3
in	in^3

Before Using the Formula

Always identify the **radius** and the **height** before applying the formula.

If the **diameter** is given, divide it by 2.

$$r = \frac{d}{2}$$

Steps to Find the Volume

Identify r and h .

Write $V = \pi r^2 h$.

Substitute the values.

Compute and write the **cubic units**.

Examples

Example 1

A cylinder has a radius of 3 cm and a height of 5 cm. Find its volume in terms of π .

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi (3)^2 (5) \\ &= 45\pi \text{ cm}^3 \end{aligned}$$

Example 2

A can has a diameter of 10 cm and a height of 7 cm. Find its volume in terms of π .

$$r = \frac{10}{2} = 5 \text{ cm}$$

$$V = \pi(5)^2(7) = 175\pi \text{ cm}^3$$

Example 3

A cylinder has a volume of $90\pi \text{ cm}^3$ and a radius of 3 cm. Find its height.

$$V = \pi r^2 h$$

$$90\pi = \pi(3)^2 h$$

$$90\pi = 9\pi h$$

$$h = 10 \text{ cm}$$

Example 4

A cylinder has a volume of $200\pi \text{ cm}^3$ and a height of 8 cm. Find its radius.

$$200\pi = \pi r^2(8)$$

$$200\pi = 8\pi r^2$$

$$r^2 = 25$$

$$r = 5 \text{ cm}$$

Example 5

The height of a cylinder is doubled while its radius stays the same. How does the volume change?

$$\begin{aligned} V_{\text{new}} &= \pi r^2(2h) \\ &= 2(\pi r^2 h) \end{aligned}$$

The new volume is 2 times the original.

Practice Exercises

Practice 1

A cylinder has a radius of 4 cm and a height of 10 cm. Find its volume in terms of π .

Practice 2

A cylindrical container has a radius of 6 cm and a height of 8 cm. Determine its volume.

Practice 3

Find the volume of a cylinder with radius 9 m and height 5 m.

Practice 4

A can has a diameter of 12 cm and a height of 15 cm. Find its volume.

Practice 5

A water tank has a radius of 3.5 m and a height of 12 m. Determine its volume. Express your answer in terms of π .

Practice 6

A cylinder has a volume of $392\pi \text{ cm}^3$ and a radius of 7 cm. Find its height.

Practice 7

A cylinder has a volume of $450\pi \text{ m}^3$ and a height of 18 m. Find its radius.

Practice 8

The radius of a cylinder is doubled while its height remains unchanged. How many times larger is the new volume than the original?

Activities

Activity 1

A cylinder has a radius of 5 cm and a height of 9 cm. Find its volume in terms of π .

Activity 2

A cylindrical container has a radius of 7 cm and a height of 6 cm. Determine its volume.

Activity 3

Find the volume of a cylinder with radius 8 m and height 11 m.

Activity 4

A can has a diameter of 16 cm and a height of 9 cm. Find its volume.

Activity 5

A storage tank has a radius of 4 m and a height of 10 m. Determine its volume. Express your answer in terms of π .

Activity 6

A cylinder has a volume of $616\pi \text{ cm}^3$ and a radius of 7 cm. Find its height.

Activity 7

A cylinder has a volume of $784\pi \text{ m}^3$ and a height of 16 m. Find its radius.

Activity 8

The radius of a cylinder is tripled while its height remains unchanged. How many times larger is the new volume than the original?

Thank you!